Sindhuja Ravikanth

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EXECUTIVE SUMMARY

AI-focused Software Engineer with 4.5+ years of experience building agentic AI systems, RAG pipelines, and full-stack cloud-native applications. Experienced in developing agents and scalable AWS architectures and leveraging transformer-based models for real-world AI applications, along with modern frontend frameworks to deliver end-to-end solutions.

SKILLS

AI/ML & Agents: LangChain, LangGraph, OpenAI SDK, RAG, Transformers, LoRA, QLoRA, Hugging Face
Programming Languages: Java, Python, JavaScript, TypeScript, C, C++, HTML, CSS
Frontend: React, Next.js, Angular, Vue.js, jQuery
Backend: Node.js, Spring Boot, Spring Framework, Hibernate
Cloud & DevOps: AWS (Lambda, S3, API Gateway, DynamoDB, Bedrock, CloudFormation), Docker, Kubernetes, Terraform, Azure
Databases: PostgreSQL, MySQL, Oracle, MongoDB, Redis
Tools & Testing: Git, Jenkins, Maven, Gradle, JUnit, Mockito, Selenium, Postman, SonarQube, Linux/Unix, CI/CD

CAREER EXPERIENCE

Software Engineer | New Jersey Department of Children and Families — September 2025 – Present

- Improved **system scalability** and **reduced infrastructure overhead by 40%** by migrating the **DCF Policy platform** to a **serverless AWS architecture** (Lambda, S3, DynamoDB) using **CloudFormation and Terraform**.
- Enhanced **document retrieval speed** by **50%** and **improved answer relevance by 30%** by building a **RAG-based policy search system using Amazon Bedrock**, enabling accurate, context-aware responses across large policy datasets.
- Reduced **manual policy lookup effort** by **60%** by developing **agentic AI workflows** using **LangGraph**, enabling **multi-step reasoning** and **structured decision support**.
- Improved **retrieval consistency** and **reduced query latency by 35%** by implementing **embedding, chunking, and indexing pipelines** for large-scale document processing.
- **Decreased manual processing time** by **50% across internal workflows** by building a **SimpliGov API integration layer**, automating form submissions and approval processes for HR and compliance teams.
- Accelerated **deployment cycles by 3x** and **reduced environment setup time by 40%** by implementing **cloud-native serverless** infrastructure with **infrastructure-as-code**.

AI/ML Engineer | Marist Poll — December 2023 – August 2025

- Increased **research team efficiency by 40%** by building a **browser-based AI agent** using **LangChain + Playwright**, orchestrating multi-step workflows for **web navigation**, **data extraction**, and **structured research synthesis**.
- Expanded **threat intelligence coverage across 2M+ URLs** by developing an **agent-driven web crawler**, enabling large-scale data extraction and automated analysis.
- Augmented the **RAG pipeline with SQL-based retrieval**, enabling **multi-source querying** across external web data and internal structured datasets.
- Built a **transformer-based text analysis** system using **Hugging Face** open-source models with **LoRA/QLoRA fine-tuning**, enabling classification, **entity extraction**, and insight generation from **50K+ social media posts**.
- Improved usability and **transparency of AI systems** by developing a **React + TypeScript interface** integrated with agent workflows, enabling **real-time interaction** and execution of AI-driven processes.

Software Developer Intern | TCPWave — May 2024 – August 2024

- **Reduced UI load time by 40%** by engineering responsive **Drupal themes using Bootstrap** and resolving 20+ SCSS styling issues.
- Accelerated **content migration efficiency** by **35%** by optimizing 50+ digital assets and implementing improved **Drupal migration workflows**.
- Improved **system reliability** to achieve a **96% test pass rate** across **150+ scenarios** by debugging **DHCP Ping Poller** and workflow issues using **Hibernate and JUnit**.

Software Developer | TCPWave **October 2021 – July 2023**

- Delivered **99.9% system uptime for 5,000+ daily users** by designing and maintaining **Java + MySQL-based** backend systems using **microservices architecture** and **design patterns**, ensuring scalability and reliability.
- Handled **1,000+ API requests daily** by developing and optimizing **30+ RESTful APIs**, enabling seamless integration across frontend applications and improving overall **system performance**.
- Improved **network visibility** by **70%** and **system observability** by **65%** by building an **IP-to-identity reporting system** integrated with **Active Directory** and **interactive monitoring dashboards**.
- Reduced **incident response time by 60%** and **accelerated release cycles by 5x** by implementing **multi-channel alerting systems** (Teams, Telegram, email) and **CI/CD pipelines** using **Jenkins, Docker,** and **SonarQube**.
- Increased **team productivity** and **user engagement** by building **CLI automation tools on** Linux and enhancing frontend applications using **Angular** and **JavaScript**, resulting in **20% efficiency gains** and **40% higher web engagement**.

PROJECTS

AI-Powered Research Workflow Agent

Reduced **research time by 60%** by building a **LangChain-based autonomous agent**, enabling structured task execution and automated knowledge synthesis through a **React interface**.

Large-Scale Cybersecurity Threat Detection System

Processed **2M+ URLs with 96% accuracy** by developing a machine learning-based **threat detection system** using **Python**, **Docker, and CI/CD pipelines**.

EDUCATION

Marist University, Poughkeepsie, NY: Master of Science in Computer Science

KMIT, Hyderabad, India: Bachelor of Technology in Computer Science and Engineering

CERTIFICATIONS

Real-Time Systems (Elite – NPTEL) - Backend system reliability, performance, and observability

Data Science for Engineers (Elite – NPTEL) - Advanced algorithms and data processing for backend systems

Social Networks (Elite Silver – NPTEL) - Scalable system architecture and distributed computing

Introduction to Research (Elite – NPTEL) - Technical troubleshooting and innovation methodologies

C Programming Language (NPTEL) - Foundation programming concepts and system-level programming